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Chapter 1: Edges

1.1 What is an Edge?

An **edge** is a location where intensity changes abruptly. Physically, edges arise from:

- *Depth discontinuity*: objects at different distances occlude one another
- *Surface orientation discontinuity*: adjacent surfaces with different normals reflect light differently
- *Reflectance discontinuity*: material or color change at an object boundary
- *Illumination discontinuity*: cast shadows create sharp intensity boundaries

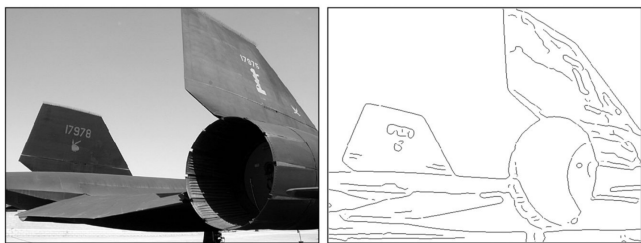


Figure 1: Visualization of Edges

Edge models. Idealised 1D profiles perpendicular to the edge:

- *Step edge*: intensity jumps instantaneously. First derivative is a spike; second derivative has a zero-crossing.
- *Ramp edge*: intensity changes linearly over several pixels (blurred step). First derivative is a plateau.
- *Roof/ridge edge*: intensity rises then falls - thin bright or dark line. Second derivative has two zero-crossings.

First-derivative operators (Sobel, Prewitt) detect step edges via large gradient magnitude. Second-derivative operators (Laplacian, LoG) detect edges via zero-crossings.

1.2 Derivative Models for Basic Edge Detection

1D analysis. For a continuous 1D signal $f(x)$, a step edge appears as:

- a *peak* in $|f'(x)|$ - the first derivative has maximum magnitude at the transition
- a *zero-crossing* in $f''(x)$ - the second derivative changes sign exactly at the edge

Discrete finite-difference approximations on a sampled signal:

$$f'(x) \approx \frac{1}{2} [f(x+1) - f(x-1)] \quad (\text{central difference, 1st order})$$

$$f''(x) \approx f(x+1) - 2f(x) + f(x-1) \quad (\text{central difference, 2nd order})$$

Two detection strategies follow:

- *First-order*: threshold $|f'(x)|$; flag pixels where gradient magnitude exceeds T .
- *Second-order*: find zero-crossings of $f''(x)$; gives sub-pixel localisation but is sensitive to noise.

2D gradient. For an image $f(x, y)$, the gradient vector

$$\nabla f = \begin{bmatrix} f_x \\ f_y \end{bmatrix} = \begin{bmatrix} \partial f / \partial x \\ \partial f / \partial y \end{bmatrix}$$

points in the direction of maximum intensity increase. Its magnitude and direction are

$$|\nabla f| = \sqrt{f_x^2 + f_y^2}, \quad \theta = \arctan\left(\frac{f_y}{f_x}\right).$$

The edge direction is *perpendicular* to ∇f (i.e. the gradient is orthogonal to the edge contour). Practical approximations replace the partial derivatives with finite-difference kernels (Sobel, Prewitt).

2D Laplacian. The isotropic second-order operator is

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \approx f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y).$$

Edge pixels are located at zero-crossings of $\nabla^2 f$. Because the Laplacian amplifies noise, it is always preceded by Gaussian smoothing (giving the LoG filter).

Key trade-off. First-order methods yield gradient direction and are easy to threshold, but produce thick edges requiring NMS. Second-order zero-crossings are precisely localised to single pixels but are noisier - hence the importance of LoG and the Canny pipeline.

1.3 Canny Edge Detector (1986)

The Canny detector is the standard pipeline for thin, well-localised, noise-robust edges. It jointly optimises three criteria: *good detection* (low miss/false-alarm rate), *good localisation* (edges marked close to true boundary), and *single response* (one edge pixel per true edge).

Algorithm.

1. *Gaussian smoothing.* Convolve with G_σ to suppress noise. Larger σ reduces noise sensitivity but blurs fine edges.
2. *Gradient computation.* Compute magnitude $M = \sqrt{G_x^2 + G_y^2}$ and direction $\theta = \arctan(G_y/G_x)$ using Sobel kernels.
3. *Non-maximum suppression (NMS).* Thin edges to one pixel. Suppress any pixel whose magnitude M is not a local maximum along the gradient direction θ . Result: single-pixel-wide ridges.
4. *Double thresholding.* Classify surviving pixels with thresholds $T_L < T_H$ (typical ratio $T_H/T_L = 2-3$):
 - $M \geq T_H$: *strong edge*
 - $T_L \leq M < T_H$: *weak edge*
 - $M < T_L$: *suppressed*
5. *Edge tracking by hysteresis.* Keep a weak edge pixel if and only if it is 8-connected to at least one strong edge pixel. Links genuine contours while discarding isolated noise responses.



Figure 2: Canny Edge detector applied to a photo

1.4 Difference of Gaussians (DoG)

DoG approximates the LoG by subtracting two Gaussians at different scales:

$$\text{DoG}_{\sigma_1, \sigma_2}(x, y) = G_{\sigma_1}(x, y) - G_{\sigma_2}(x, y), \quad \sigma_1 < \sigma_2.$$

When $\sigma_2/\sigma_1 \approx 1.6$, DoG closely approximates $(\sigma_2^2 - \sigma_1^2) \nabla^2 G_\sigma$. Edge locations are zero-crossings of the DoG response. Computationally cheaper than LoG (two Gaussian blurs, no explicit Laplacian kernel). Used as the blob/edge detector in the SIFT feature pipeline.

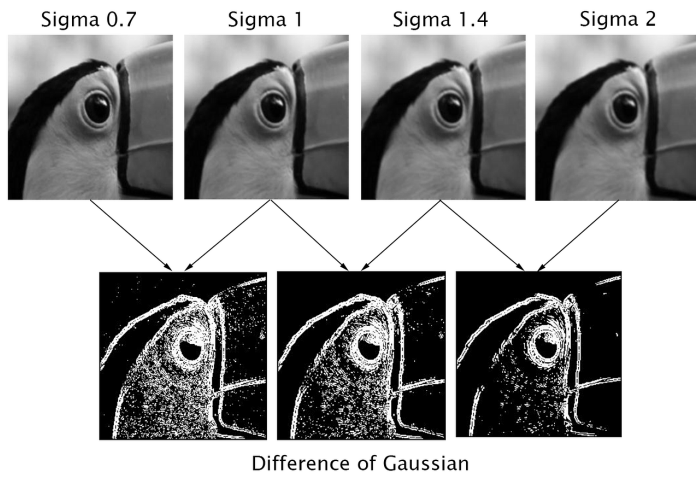


Figure 3: Difference of Gaussian Illustration

Chapter 2: Corners

2.1 Corner Detection

A **corner** is a point with large intensity variation in *multiple* directions simultaneously. Unlike edges (localised in only one direction), corners are well-localised in 2D and are repeatable under viewpoint and illumination changes.

- **Saliency:** Corners are found at “interesting” regions of an image.
- **Localization:** Corners occupy a small region: used to pinpoint information.
- **Repeatability:** Photometric or geometric transformations do not affect the corner.

Flat vs. edge vs. corner. Shift a small window W by $(\Delta x, \Delta y)$; the sum of squared differences is

$$E(\Delta x, \Delta y) = \sum_{(x,y) \in W} [f(x + \Delta x, y + \Delta y) - f(x, y)]^2.$$

- *Flat:* E small for all shift directions
- *Edge:* E large only perpendicular to the edge
- *Corner:* E large in all directions

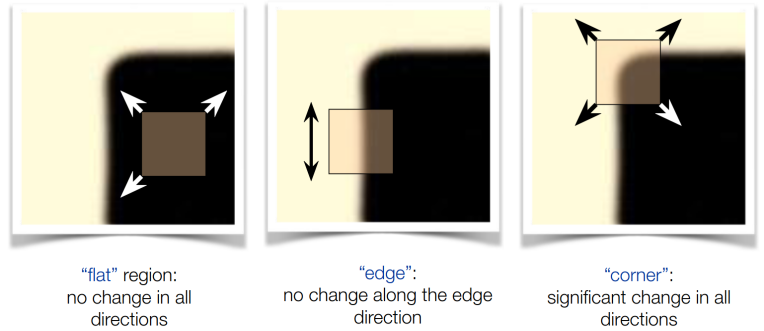


Figure 4: Flat vs. edge vs. corner.

2.2 Moravec Corner Detector (1980)

The earliest interest-point detector (Moravec, 1980). The core idea: a corner is where shifting a small window in *any* direction causes a large intensity change.

Algorithm.

1. Fix a window W (typically 5×5) centred at pixel (x, y) .
2. Compute the SSD for each of four discrete shifts $\Delta \in \{(1, 0), (0, 1), (1, 1), (1, -1)\}$:

$$\text{SSD}_{\Delta}(x, y) = \sum_{(s,t) \in W} [f(x+s+\Delta x, y+t+\Delta y) - f(x+s, y+t)]^2.$$

3. Take the *minimum* over all four shifts as the corner strength:

$$C(x, y) = \min_{\Delta} \text{SSD}_{\Delta}(x, y).$$

The minimum captures the weakest direction - a corner must be strong in *every* direction.

4. Threshold: candidate corner if $C(x, y) > T$.
5. Non-maximum suppression: keep only local maxima of C in a neighbourhood.

Interpretation of C .

- C large in all directions $\Rightarrow$ corner
- C large only perpendicular to one direction $\Rightarrow$ edge
- C small everywhere $\Rightarrow$ flat region

Limitations.

- Only 4 discrete shift directions - anisotropic; misses edges at intermediate angles.
- Uniform (box) window weighting - no noise suppression; pixels near the boundary contribute as much as those at the centre.
- Hard threshold on C ; no smooth response grading.
- Not rotation-invariant.

Harris addresses all four: the structure tensor M integrates over *all* directions via gradients, the Gaussian window downweights boundary pixels, the response R is continuous, and M 's eigenvalues are rotation-invariant.

2.3 Harris Corner Detector (1988)

Harris & Stephens (1988) fix Moravec's limitations by replacing discrete shifts with a continuous gradient formulation.

Algorithm.

1. *Compute image gradients.* Compute $f_x = \partial f / \partial x$ and $f_y = \partial f / \partial y$ using Sobel, Prewitt (or similar) kernels.
2. *Form the structure tensor.* At each pixel, build the outer-product matrix and smooth it with a Gaussian window G_σ :

$$M(x,y) = G_\sigma * \begin{bmatrix} f_x^2 & f_x f_y \\ f_x f_y & f_y^2 \end{bmatrix}.$$

This integrates gradient information over a neighbourhood, weighting central pixels more heavily. The Taylor-linearised SSD then satisfies

$$E(\Delta x, \Delta y) \approx \begin{bmatrix} \Delta x & \Delta y \end{bmatrix} M \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}.$$

3. *Classify via eigenvalues.* The eigenvalues $\lambda_1 \geq \lambda_2$ of M describe local geometry:

λ_1	λ_2	Region type
small	small	flat
large	small	edge
large	large	corner

4. *Compute corner response.* To avoid eigendecomposition, use

$R = \det(M) - k [\text{tr}(M)]^2 = \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2)^2, \quad k \approx 0.04\text{--}0.06.$
 - $R \gg 0$: corner (both λ large)
 - $R \ll 0$: edge (one λ dominates)
 - $|R| \approx 0$: flat
5. *Threshold and NMS.* Keep pixels where $R > T$ and R is a local maximum in a 3×3 (or larger) neighbourhood.

Properties. Rotation-invariant (eigenvalues are invariant to rotation of image coordinates). Not scale-invariant - the detected corners change as the image is rescaled.

2.4 Shi-Tomasi Corner Detector (1994)

Shi and Tomasi replace the Harris response with

$$R_{\text{ST}} = \min(\lambda_1, \lambda_2).$$

A pixel is a corner if $\min(\lambda_1, \lambda_2) > \tau$. Both eigenvalues must exceed the threshold, giving more uniformly distributed, stable corners. Used in OpenCV's `goodFeaturesToTrack`.

2.5 Noble Corner Detector (1998)

Noble replace the Harris response with

$$R_{\text{ST}} = \frac{\det(M)}{\text{trace}(M) + \epsilon}$$

where ϵ is a small positive constant.

2.6 Förstner Corner Detector (1987)

Förstner & Gülch (1987) predates Harris and takes a fundamentally different approach: rather than collapsing the structure tensor into a single scalar

response, it estimates an *optimal sub-pixel location* for each corner candidate and characterises the uncertainty of that location with an *error ellipse*.

Motivation. Each image gradient $\nabla f(x_i, y_i) = (f_{x,i}, f_{y,i})$ defines a constraint line through (x_i, y_i) orthogonal to the gradient: all points along the edge satisfy this line equation. At a true corner, many such lines from different directions meet at a single point. The optimal intersection $\hat{\mathbf{p}} = (\hat{u}, \hat{v})$ minimises the weighted sum of squared perpendicular distances to all gradient lines in a neighborhood W :

$$\hat{\mathbf{p}} = \arg \min_{\mathbf{p}} \sum_{(x_i, y_i) \in W} w_i (f_{x,i}(u - x_i) + f_{y,i}(v - y_i))^2.$$

Setting the gradient to zero gives the normal equations $M\hat{\mathbf{p}} = \mathbf{b}$, where M is the (Gaussian-windowed) structure tensor and

$$\mathbf{b} = G_\sigma * \begin{bmatrix} f_x(f_x x + f_y y) \\ f_y(f_x x + f_y y) \end{bmatrix}.$$

The sub-pixel corner estimate is therefore $\hat{\mathbf{p}} = M^{-1}\mathbf{b}$.

Error ellipse. The covariance of $\hat{\mathbf{p}}$ (assuming i.i.d. gradient noise with variance σ_0^2) is

$$\Sigma_{\hat{\mathbf{p}}} = \sigma_0^2 M^{-1}.$$

Its shape is an ellipse whose axes align with the eigenvectors of M and whose semi-axis lengths are proportional to $1/\sqrt{\lambda_i}$. A small, circular ellipse means the corner is well-localised in all directions - the hallmark of a true corner.

Two quality measures. With eigenvalues $\lambda_1 \geq \lambda_2$ of M :

- **Roundness** q - how circular the error ellipse is:

$q = \frac{4 \det M}{(\text{tr } M)^2} = \frac{4\lambda_1 \lambda_2}{(\lambda_1 + \lambda_2)^2} \in (0, 1].$

$q \rightarrow 1$: ellipse is nearly circular (good corner); $q \rightarrow 0$: highly elongated (edge - uncertain along it).
- **Size** (localization precision) w - the harmonic mean of the eigenvalues:

$w = \frac{\det M}{\text{tr } M} = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}.$

Large w : small error ellipse (well-localised); small w : large ellipse (flat or noisy region).

Algorithm.

1. Compute Gaussian-windowed structure tensor M and right-hand side $\mathbf{b}$ at every pixel.
2. Compute q and w from M .
3. A pixel is a corner candidate if $q > q_0$ (typically 0.5) *and* $w > w_0$ (data-dependent threshold).
4. Sub-pixel refinement: compute $\hat{\mathbf{p}} = M^{-1}\mathbf{b}$ at each candidate.
5. NMS on w to suppress non-maximal candidates.

Key distinction from Harris. Harris fuses roundness and size into a single response $R = \det M - k(\text{tr } M)^2$, which can pass the threshold via a large determinant even if the ellipse is elongated (q small). Förstner's two-threshold scheme (q and w independently) avoids false-positives on strong edges, and the sub-pixel estimate $\hat{\mathbf{p}}$ gives localisation accuracy beyond the pixel grid.

2.7 The SUSAN Family (1997)

SUSAN (Smallest Univalue Segment Assimilating Nucleus) (Smith & Brady, 1997) detects edges and corners without computing image derivatives, using only intensity comparisons within a circular mask.

Core idea: the USAN area. For each pixel (the *nucleus*) $\mathbf{r}_0$, place a circular mask of radius ≈ 3.4 pixels (37 pixels). Count how many mask pixels have similar intensity to the nucleus:

$$n(\mathbf{r}_0) = \sum_{\mathbf{r} \in \text{mask}} c(\mathbf{r}, \mathbf{r}_0),$$

using the comparison function

$$c(\mathbf{r}, \mathbf{r}_0) = e^{-\left(\frac{f(\mathbf{r}) - f(\mathbf{r}_0)}{t}\right)^6},$$

where t is the intensity similarity threshold. (A hard binary version $c = \mathbf{1}[|f(\mathbf{r}) - f(\mathbf{r}_0)| \leq t]$ is also common.) The set of similar pixels is the **USAN**.

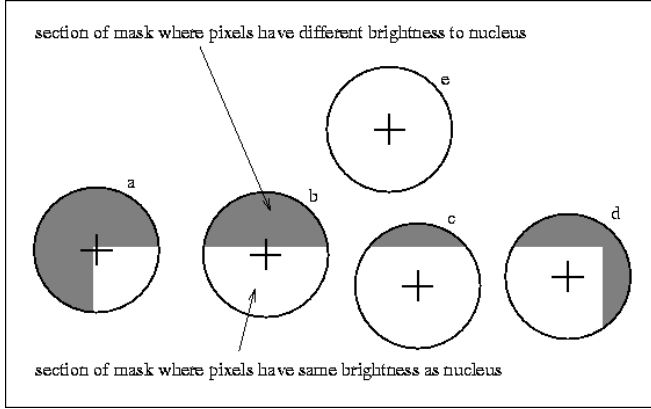
Geometric interpretation of n .

- Nucleus in a *flat region*: nearly all mask pixels are similar $\Rightarrow n \approx n_{\max}$ (large)
- Nucleus on an *edge*: roughly half the mask crosses the boundary $\Rightarrow n \approx n_{\max}/2$
- Nucleus at a *corner*: only a small wedge of the mask is similar $\Rightarrow n \ll n_{\max}/2$ (small)

Corner response. Define the geometric threshold $g = 3n_{\max}/4$. Then:

$$R(\mathbf{r}_0) = \begin{cases} g - n(\mathbf{r}_0), & n(\mathbf{r}_0) < g, \\ 0, & \text{otherwise.} \end{cases}$$

$R > 0$ only where the USAN is smaller than g , i.e. at edges and corners. Corners give the largest R . Apply NMS to retain local maxima.



Edge detection with SUSAN. Edges are where $n \approx n_{\max}/2$. The edge direction is estimated from the centroid of the USAN relative to the nucleus.

Properties.

- No image derivatives - robust to noise without an explicit smoothing step; t controls sensitivity vs. noise.
- A single circular mask replaces the rectangular window of Moravec and the gradient kernels of Harris.
- Not rotation-invariant (the discrete circular mask breaks symmetry slightly) and not scale-invariant.
- Directly inspired FAST: FAST's segment test is a fast approximation of the USAN area, replacing the full 37-pixel count with a 16-pixel ring and a contiguous-arc criterion.

2.8 The FAST Corner Detector (2006)

Features from Accelerated Segment Test (Rosten & Drummond, 2006). Designed for real-time applications where Harris/Shi-Tomasi are too slow. All decisions are intensity comparisons - no gradients, no matrix operations.

Circle specification. Consider the 16 pixels $p_1, \dots, p_{16}$ on a Bresenham circle of radius 3 around centre pixel p , labelled clockwise from the top. Their offsets from p are:

$$\begin{aligned} &(-3, 0), (-3, 1), (-2, 2), (-1, 3), (0, 3), (1, 3), (2, 2), (3, 1), \\ &(3, 0), (3, -1), (2, -2), (1, -3), (0, -3), (-1, -3), (-2, -2), (-3, -1). \end{aligned}$$

Segment test. Let I_p be the intensity of p and t a threshold. Each circle pixel p_i is classified as:

- *brighter*: $I_{p_i} > I_p + t$
- *darker*: $I_{p_i} < I_p - t$
- *similar*: otherwise

p is declared a corner if there exist N *contiguous* circle pixels all brighter, or all darker, than the respective threshold. Common variants: **FAST-9** ($N = 9$) and **FAST-12** ($N = 12$). FAST-12 is more conservative (fewer detections, higher precision); FAST-9 detects more corners.

High-speed rejection (FAST-12). Test only the four cardinal pixels p_1, p_5, p_9, p_{13} (spaced 90° apart). For 12 contiguous pixels to exist, at least

3 of these 4 must all be brighter or all darker. If this condition fails, p is immediately rejected - at most 4 memory accesses. This pre-test eliminates $\approx 75\%$ of pixels in typical images.

Corner score and NMS. After the segment test, assign a score

$$V(p) = \max \left(\sum_{p_i \in S_b} (I_{p_i} - I_p - t), \sum_{p_i \in S_d} (I_p - I_{p_i} - t) \right),$$

where S_b (resp. S_d) is the set of brighter (resp. darker) arc pixels. Retain only local maxima of V - adjacent corners collapse to the single strongest response.

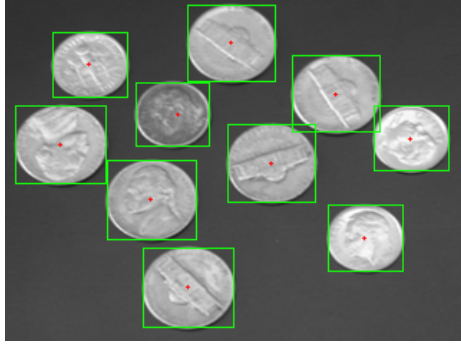
Properties.

- No gradient computation, no matrix inversion - extremely fast; suitable for real-time SLAM, tracking, and mobile AR.
- Not rotation-invariant (circle is fixed; no orientation assigned). **ORB** extends FAST by adding an orientation via the intensity centroid method and pairing it with a rotated BRIEF descriptor.
- Not scale-invariant - run at multiple scales (image pyramid) to gain partial scale robustness.
- Sensitive to noise: no smoothing step. Tuning t balances sensitivity vs. false positives.

Detector	Year	Rotation inv.	Scale inv.	Speed
Moravec	1980	No	No	Fast
Harris	1988	Yes	No	Moderate
Shi-Tomasi	1994	Yes	No	Moderate
FAST	2006	No	No	Very fast

Chapter 3: Blobs

A **blob** is a connected image region whose intensity (or color) is approximately uniform and differs from its surroundings. Unlike edges (1D features) or corners (0D point features), blobs are 2D structures characterised by both *position* and *scale*. The central challenge is detecting blobs across a range of sizes without prior knowledge of which scale to look at.



3.1 Scale-Space and Characteristic Scale

Scale-space. Convolving an image f with a Gaussian of width σ gives the linear scale-space:

$$L(x, y; \sigma) = G_\sigma(x, y) * f(x, y).$$

Increasing σ progressively suppresses fine detail while leaving coarser structure intact. The full set $\{L(\cdot; \sigma)\}_{\sigma>0}$ is the *Gaussian scale-space* of f .

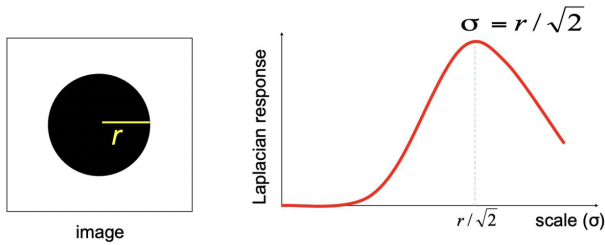
Why the (scale-normalised) LoG? The Laplacian $\nabla^2 L$ measures how much a pixel differs from its neighborhood - large for blob centers, near zero on flat regions. However, raw LoG responses shrink as σ grows (the Gaussian spreads and flattens). To compare responses across scales fairly, multiply by σ^2 :

$$\mathcal{L}(x, y; \sigma) = \sigma^2 \nabla^2 L(x, y; \sigma) = \sigma^2 (\nabla^2 G_\sigma) * f(x, y).$$

This is the *scale-normalised LoG* (Lindeberg, 1994). For a bright circular blob of radius r , $\mathcal{L}$ peaks at the *characteristic scale*:

$$\sigma^* = \frac{r}{\sqrt{2}},$$

so the blob radius can be read off as $r = \sigma^* \sqrt{2}$. Detecting blobs therefore reduces to finding local extrema of $\mathcal{L}$ jointly over space *and* scale.



3.2 LoG Blob Detector

Algorithm.

1. Build a discrete scale-space by computing $L(\cdot; \sigma_k)$ for $k = 1, \dots, K$ (typically spaced by a factor of $\sqrt{2}$ or 2).
2. At each scale compute the scale-normalised LoG: $\mathcal{L}(x, y; \sigma_k) = \sigma_k^2 \nabla^2 G_{\sigma_k} * f$.
3. Detect *local extrema* in the 3D volume (x, y, σ) : a pixel is a blob candidate if $\mathcal{L}$ is larger (or smaller) than all 26 neighbors - its 3×3 spatial neighborhood at the same scale *and* the corresponding 3×3 patches at the adjacent scales.
4. Threshold to suppress weak responses; NMS to retain strongest detections.

The result is a set of circles (x_i, y_i, r_i) - position and estimated radius - that are scale-covariant: if the image is rescaled by a factor s , the detected radii scale by the same s .

3.3 Difference of Gaussians (DoG) Blob Detector

DoG approximates the scale-normalised LoG and is the basis of SIFT blob detection.

Approximation. For adjacent scales σ and $k\sigma$ (with $k > 1$ close to 1):

$$G_{k\sigma} - G_\sigma \approx (k-1)\sigma \frac{\partial G_\sigma}{\partial \sigma} = (k-1)\sigma^2 \nabla^2 G_\sigma,$$

so

$$\text{DoG}(x, y; \sigma) = L(x, y; k\sigma) - L(x, y; \sigma) \approx (k-1)\mathcal{L}(x, y; \sigma).$$

The constant $(k-1)$ is irrelevant for locating extrema. With $k = 2^{1/s}$ (s intervals per octave), SIFT typically uses $s = 3$, computing $s+3$ blurred images per octave (and $s+2$ DoG images), then detecting extrema across s interior DoG scales per octave.

Scale-space pyramid.

1. Convolve f with Gaussians $G_\sigma, G_{k\sigma}, G_{k^2\sigma}, \dots$ (one octave).
2. Subtract adjacent pairs to get DoG images.
3. Detect local extrema in the DoG stack as above.
4. Downsample by $2\times$ to start the next octave.

Two Gaussian blurs and one subtraction replace one Laplacian convolution, at roughly the same cost; the pyramid additionally provides free multi-scale coverage.

3.4 Hessian-Based Blob Detection

The Hessian matrix at scale σ captures second-order intensity structure:

$$\mathbf{H}(x, y; \sigma) = \sigma^2 \begin{bmatrix} L_{xx} & L_{xy} \\ L_{xy} & L_{yy} \end{bmatrix},$$

where $L_{xx} = \partial^2 L / \partial x^2$, etc. (scale-normalised second derivatives). For a circular blob:

- $\det(\mathbf{H}) = \sigma^4 (L_{xx}L_{yy} - L_{xy}^2)$ is maximised at the blob center and characteristic scale. Used in **SURF** (Bay et al., 2006).
- $\text{tr}(\mathbf{H}) = \sigma^2 (L_{xx} + L_{yy}) = \mathcal{L}$ - the LoG. The trace and determinant share the same extrema for isotropic blobs.

SURF (Speeded Up Robust Features) approximates the Gaussian second derivatives L_{xx}, L_{yy}, L_{xy} with box filters, enabling $O(1)$ evaluation via *integral images*:

$$II(x, y) = \sum_{x' \leq x, y' \leq y} f(x', y').$$

Any rectangular sum is computed in 4 table lookups. Box-filter approximations allow SURF to evaluate the Hessian determinant at any scale and position in constant time, making it significantly faster than LoG or DoG.

3.5 Maximally Stable Extremal Regions (MSER)

MSER (Matas et al., 2004) detects blob-like regions without any Gaussian filtering, using only intensity thresholding.

Extremal regions. For a threshold t , the set $\mathcal{R}(t) = \{(x, y) : f(x, y) \leq t\}$ is thresholded; each connected component of $\mathcal{R}(t)$ is an *extremal region*. As t increases from 0 to $L-1$, regions grow and merge (a sequence tracked efficiently by union-find).

Stability criterion. A region $\mathcal{R}(t)$ is *maximally stable* if its relative area change

$$q(t) = \frac{|\mathcal{R}(t+\Delta) \setminus \mathcal{R}(t-\Delta)|}{|\mathcal{R}(t)|}$$

is a local minimum over t . Stable regions are those that persist over a wide range of thresholds - they correspond to distinctly darker (or lighter) blobs with clear boundaries.

Properties.

- Invariant to monotone intensity transformations (re-thresholding gives the same regions).
- *Affine covariant*, not just similarity covariant - MSER shapes are fit with ellipses and normalised to a canonical circular patch, enabling wide-baseline matching across significant viewpoint change.
- No smoothing step; very fast (near-linear in pixels).
- Fails on textured or blurry boundaries where no stable threshold exists.

3.6 Moment-Based Blob Detection

The scale-space detectors search for extrema of a filter response; the moment-based approach instead *segments first* - by thresholding, background subtraction, or an MSER/connected-component sweep - and then characterises each candidate region by integrating over it. Moments turn a region into a handful of numbers describing its mass, position, spread, and orientation.

Raw moments. For a region $\mathcal{R}$ with weight $f(x, y)$ (the binary mask, or the intensity itself for a grey-level blob):

$$m_{pq} = \sum_{(x,y) \in \mathcal{R}} x^p y^q f(x, y).$$

The zeroth moment m_{00} is the area (or total mass), and the first moments give the centroid:

$$(\bar{x}, \bar{y}) = \left(\frac{m_{10}}{m_{00}}, \frac{m_{01}}{m_{00}} \right),$$

which localises the blob to sub-pixel accuracy - unlike the integer-pixel extremum returned by LoG or DoG.

Central moments. Subtracting the centroid makes the moments translation invariant:

$$\mu_{pq} = \sum_{(x,y) \in \mathcal{R}} (x - \bar{x})^p (y - \bar{y})^q f(x, y).$$

Shape from second-order moments. The normalised second-order central moments form the region’s covariance matrix:

$$\mathbf{C} = \frac{1}{\mu_{00}} \begin{bmatrix} \mu_{20} & \mu_{11} \\ \mu_{11} & \mu_{02} \end{bmatrix}.$$

Its eigenvectors and eigenvalues $\lambda_1 \geq \lambda_2$ define the *equivalent ellipse* - the uniform ellipse with the same second moments as the blob:

- semi-axes $a = 2\sqrt{\lambda_1}$, $b = 2\sqrt{\lambda_2}$, so a disc of radius r gives $\lambda_1 = \lambda_2 = r^2/4$ and the *scale is read off directly* - $r = 2\sqrt{\lambda}$ - with no scale-space search;
- orientation $\theta = \frac{1}{2} \arctan \frac{2\mu_{11}}{\mu_{20} - \mu_{02}}$, giving each blob a canonical frame for descriptor extraction;
- eccentricity $e = \sqrt{1 - \lambda_2/\lambda_1}$, which measures how elongated the blob is. This ellipse fit is exactly the normalisation step used to make MSER regions affine covariant.

Scale and rotation invariance. Dividing out the area makes central moments scale invariant:

$$\eta_{pq} = \frac{\mu_{pq}}{\mu_{00}^{(p+q)/2+1}},$$

and particular combinations of the η_{pq} are additionally rotation invariant - the **Hu moments** (Hu, 1962), of which the first two are

$$\phi_1 = \eta_{20} + \eta_{02}, \quad \phi_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2.$$

The seven Hu invariants (the seventh also flips sign under reflection) serve as a compact shape descriptor for matching or classifying detected blobs.

Filtering candidates. Segmentation over-produces regions, so moment-derived shape statistics are used to reject them. The standard criteria (as in OpenCV’s `SimpleBlobDetector`) are area μ_{00} , circularity $4\pi\mu_{00}/P^2$ for perimeter P , inertia ratio λ_2/λ_1 , and convexity (region area over convex-hull area). A blob survives if all of them fall in their accepted range.

Trade-offs. Moments are one $O(N)$ pass over the region and need no filter bank, and they yield sub-pixel position, orientation, and elongation for free - which is why they dominate in controlled settings (microscopy, industrial inspection, marker and centroid tracking). The weakness is the dependence on a prior segmentation: results are only as good as the threshold, and the method degrades on cluttered or low-contrast scenes where no single threshold

isolates the blob. This is the inverse of the LoG/DoG trade-off, which needs no segmentation but must search scale explicitly.

3.7 Other Blob Detectors

- **Radial symmetry.** The fast radial symmetry transform (Loy and Zelinsky, 2003) has every gradient vote for the point a distance n ahead of it; points accumulating many votes across several radii n are radially symmetric blob centers. Gradient-driven rather than intensity-driven, so it survives uneven illumination - the standard choice for eyes, pupils, and circular fiducials.
- **Watershed.** Treat $-f$ as a terrain and flood from local minima; catchment basins are blobs and the watershed lines their boundaries. Marker-controlled flooding (seeding only from chosen minima) is what keeps it from over-segmenting, and is the usual way to split touching blobs that thresholding merges into one region.
- **Affine-covariant extensions.** Harris-affine and Hessian-affine (Mikolajczyk and Schmid, 2004) start from a scale-space detection, then iterate the second-moment matrix to normalise the patch to a circle. This makes the detection covariant to affine - not just similarity - transforms, at substantially higher cost.

Detector	Basis	Scale inv.	Speed
LoG	Scale-norm. Laplacian	Yes	Slow
DoG (SIFT)	Gaussian subtraction	Yes	Moderate
SURF	Box-filter Hessian	Yes	Fast
MSER	Threshold sweep	Yes (affine)	Fast
Moments	Region integrals	Yes (from λ)	Fast
Radial sym.	Gradient voting	Yes (radii set)	Moderate

Chapter 4: Features

4.1 Feature Pipeline

A **feature** is a compact, meaningful representation of local image structure that can be reliably detected and matched across different images of the same scene. Features bridge raw pixels and high-level tasks such as object recognition, 3D reconstruction, and image retrieval.



A complete feature pipeline has three stages:

- **Detector** - locates *where* interesting structure exists (position, scale, orientation). A good detector is *repeatable*: it fires at the same physical scene point across different images. Flat regions and edges are poor candidates; corners and blobs are preferred because they are localised in multiple directions (see CORNERS and BLOBS).
- **Descriptor** - encodes *what* the local patch looks like into a fixed-length vector or bit string. It must be *distinctive* (different points $\rightarrow$ different vectors) and *invariant* (the same point under different conditions $\rightarrow$ the same vector).
- **Matcher** - establishes correspondences by comparing descriptors, then uses geometric verification (e.g. RANSAC) to filter outliers.

Desiderata. Repeatability (detector fires at the same scene point despite appearance change); distinctiveness (correct match much closer in descriptor space than any incorrect one); locality (small patch $\Rightarrow$ robust to occlusion); sufficient quantity and computational efficiency for the application.

Invariance vs. discriminability. Baking invariance to a transformation into a descriptor always discards information - the art is choosing which invariances matter (scale, rotation, illumination) while preserving enough structure to remain discriminative.

Taxonomy of this section.

1. *Local keypoint pipelines* (SIFT, SURF, binary): detect sparse interest points; describe each with a local patch vector.
2. *Dense descriptors* (HOG): compute descriptors on a regular grid over a detection window, not at sparse keypoints - used for classification and detection.
3. *Matching and geometric verification*: compare descriptor sets across images; filter with ratio test and RANSAC.
4. *Image-level aggregation* (BoW, VLAD, Fisher): encode an entire image's descriptor set into one fixed-length vector for retrieval and classification.

4.2 RANSAC

RANdom SAMple Consensus (RANSAC, Fischler & Bolles 1981) is a general-purpose iterative algorithm for robustly fitting a parametric model to data that contains a large fraction of *outliers* - observations that do not conform to the model. Classical estimators such as least squares are *non-robust*: a single outlier can arbitrarily corrupt the solution. RANSAC instead searches for the model supported by the *largest consistent subset* of the data, tolerating outlier fractions ϵ well above 50%.

4.2.1 Generic RANSAC

Motivating example. Suppose you have 100 2D points that roughly lie on a straight line, but 60 of them are random noise (outliers). Least squares fits

the line to *all* 100 points and gets pulled toward the noise. RANSAC instead asks: “*which line has the most points near it?*” It tries many candidate lines, each built from a tiny random sample, and picks the winner.



Parametric model family. A *parametric model* is any mathematical structure described by a fixed set of numbers (parameters) θ . The model family $\mathcal{M}(\theta)$ is the collection of all possible instances as θ varies:

- A 2D line $ax + by + c = 0$ has parameters $\theta = (a, b, c)$.
- A circle $(x - h)^2 + (y - k)^2 = r^2$ has parameters $\theta = (h, k, r)$.
- A 2D affine transform has 6 parameters; a homography has 8 (up to scale).

The model family fixes the *shape* of what you are fitting; the parameters pin down the specific instance.

Residual function. The residual $\rho(p_i, \theta)$ measures *how far* a data point p_i is from the model $\mathcal{M}(\theta)$ - i.e. how badly it violates the model. A small residual means p_i is consistent with θ ; a large residual means it is an outlier. The exact form depends on the model:

- *Line*: perpendicular (point-to-line) distance.

$$\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$$

- *Circle*: $|\sqrt{(x_i - h)^2 + (y_i - k)^2} - r|$ (how far the point is from the circle boundary).
- *Homography*: pixel reprojection error $\|\mathbf{x}'_i - H\mathbf{x}_i\|$ (how far the predicted location is from the observed one).

Inlier threshold ϵ . A point p_i is declared an *inlier* if $\rho(p_i, \theta) < \epsilon$ - its residual is small enough to plausibly arise from measurement noise rather than the point being wrong. Choosing ϵ is the main tuning knob: too tight and genuine matches are rejected; too loose and outliers are accepted. A principled choice is $\epsilon \approx 2.5\sigma$ where σ is the expected noise standard deviation (covers 95% of the Gaussian tail).

Minimum sample size s . Every model requires a minimum number of points to be determined uniquely. Fewer than s points leave the model underdetermined; exactly s gives a unique solution (or finitely many). RANSAC always draws exactly s points per hypothesis to keep each trial as fast as possible and to minimise the chance of including an outlier.

Algorithm.

1. **Sample.** Draw s points uniformly at random from $\mathcal{D} = \{p_i\}_{i=1}^n$ (without replacement).
2. **Fit.** Solve for parameters $\hat{\theta}$ using only those s points.
3. **Score.** Count all data points whose residual falls below ϵ - these are the inliers for this hypothesis:

$$\mathcal{I}(\hat{\theta}) = \{i : \rho(p_i, \hat{\theta}) < \epsilon\}.$$

4. **Iterate.** Repeat steps 1-3 for k iterations; keep the hypothesis θ^* with the largest inlier count $|\mathcal{I}|$.
5. **Refit.** Re-estimate θ^* using *all* inliers $\mathcal{I}(\theta^*)$ via least squares. Because inliers are (mostly) noise-free true points, this final fit is far more accurate than any single s -point hypothesis.

Number of iterations. Each trial succeeds (all s drawn points are inliers) with probability $(1 - e)^s$, where e is the outlier fraction. The probability that at least one of k independent trials succeeds is $1 - (1 - (1 - e)^s)^k$. Setting this equal to p :

$$k = \frac{\log(1 - p)}{\log(1 - (1 - e)^s)}.$$

Since e is unknown, it is estimated adaptively: after each iteration, $e \leftarrow 1 - |T^*|/n$ and k is updated, enabling early termination as soon as a good hypothesis is found.

Examples (model / s / residual).

- *2D line* ($s = 2$): perpendicular point-to-line distance.
- *Circle* ($s = 3$): distance from point to circle boundary.
- *3D plane* ($s = 3$): point-to-plane distance (LiDAR ground removal).
- *Homography / affine / fundamental matrix*: transfer error in pixels - see below.

Properties and limitations.

- *Probabilistic guarantee*: succeeds with probability $\geq p$; outcomes vary across runs.
- *Iteration cost*: k grows exponentially with e and s . For the fundamental matrix ($s = 8$) at $e = 0.6$, $p = 0.99$, this exceeds 10^4 iterations.
- *Degenerate samples*: some s -point draws are geometrically degenerate (e.g. three collinear points cannot determine a unique homography). Discard and resample.

4.2.2 RANSAC for Geometric Verification (Features)

In the feature-matching pipeline, raw descriptor matches contain many *outliers* (incorrect correspondences). RANSAC robustly estimates a geometric model - homography H , affine A , or fundamental matrix F - from noisy correspondences $\{(\mathbf{x}_i, \mathbf{x}'_i)\}$. Here data points are correspondence pairs, the model is a transform matrix, and the residual is a *transfer error* in pixel space.

Algorithm (feature-specific instantiation).

1. **Sample.** Randomly draw the minimal set of correspondences: $s = 3$ (affine), $s = 4$ (homography), $s = 8$ (fundamental matrix).
2. **Fit.** Estimate M from the sample via DLT or the normal equations.
3. **Score.** Count inliers with transfer error $d(\mathbf{x}'_i, M\mathbf{x}_i) < \epsilon$ (typically $\epsilon = 3-5$ px).
4. **Iterate** for k iterations; keep M^* with the most inliers.
5. **Refit** M^* to all inliers $\mathcal{I}(M^*)$ via least squares.

Transfer error metrics.

- *Symmetric reprojection error* (homography): $d = d(\mathbf{x}', H\mathbf{x})^2 + d(\mathbf{x}, H^{-1}\mathbf{x}')^2$.
- *Sampson distance* (fundamental matrix): first-order approximation to geometric reprojection error, cheaper than the exact epipolar distance.

Typical iteration counts for $e = 0.5$, $p = 0.99$:

Model	Min. sample s	Iterations k
Line (2D)	2	17
Affine	3	35
Homography	4	72
Fundamental matrix	8	1177

Notable variants.

- **MSAC** (Torr & Zisserman 2000): truncated ℓ_2 loss instead of binary inlier count - weights near-inliers more gracefully.
- **MLESAC** (Torr & Zisserman 2000): maximises likelihood under an inlier-Gaussian / outlier-uniform mixture (EM on the inlier set).
- **PROSAC** (Chum & Matas 2005): samples progressively from the most confident matches (ranked by descriptor distance), reducing iterations by orders of magnitude.
- **LO-RANSAC** (Chum et al. 2003): after each best hypothesis, runs a local optimisation on the inlier set before the global update, greatly improving solution quality.

Chapter 5: Feature Detection Algorithms

5.1 SIFT (Lowe, 1999/2004)

Scale-Invariant Feature Transform is the canonical local feature pipeline. It is invariant to scale, rotation, and moderate affine distortion, and robust to illumination change. The algorithm has four stages.

1. Scale-space extrema detection.

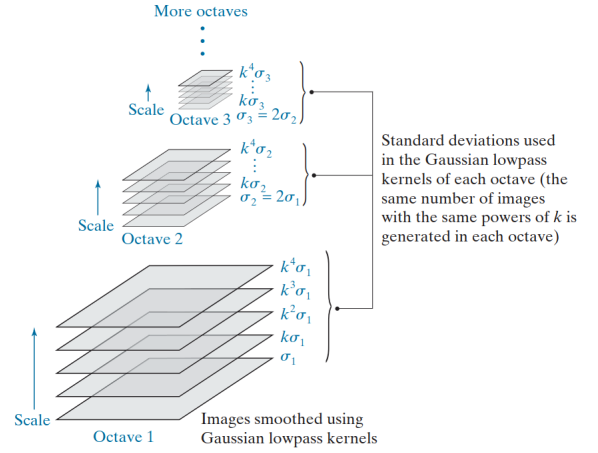
Build a Gaussian scale-space by convolving the image with Gaussians at increasing σ :

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y), \quad G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/(2\sigma^2)}.$$

Group scales into *octaves*: within each octave use s intervals, so the scale ratio between adjacent levels is $k = 2^{1/s}$ (Lowe uses $s = 3$). The base scale of each octave is $\sigma_0 = 1.6$ (at the first octave; the image is upsampled $2\times$ before the first octave in the reference implementation). Compute the *Difference of Gaussians* (DoGs):

$$D(x, y, \sigma) = L(x, y, k\sigma) - L(x, y, \sigma) \approx (k - 1) \sigma^2 \nabla^2 L.$$

Each octave produces $s + 3$ smoothed images and $s + 2$ DoG images; extrema are detected in the s interior DoG scales (leaving one margin on each side for neighbor comparison). A pixel is a candidate if $D(x, y, \sigma)$ is larger (or smaller) than all 26 neighbors in the $3 \times 3 \times 3$ (x, y, σ) volume.



2. Keypoint localisation (sub-pixel / sub-scale).

Fit a 3D quadratic to D near each candidate by Taylor expansion about the candidate location $\mathbf{x}_0 = (x, y, \sigma)^T$:

$$D(\mathbf{x}_0 + \hat{\mathbf{x}}) \approx D(\mathbf{x}_0) + \frac{\partial D}{\partial \mathbf{x}} \hat{\mathbf{x}} + \frac{1}{2} \hat{\mathbf{x}}^T \frac{\partial^2 D}{\partial \mathbf{x}^2} \hat{\mathbf{x}}.$$

Setting the derivative to zero gives the interpolated offset:

$$\hat{\mathbf{x}} = - \left(\frac{\partial^2 D}{\partial \mathbf{x}^2} \right)^{-1} \frac{\partial D}{\partial \mathbf{x}}.$$

The Hessian and gradient are approximated by finite differences, e.g.

$$\frac{\partial D}{\partial x} \approx \frac{D(x+1, y, \sigma) - D(x-1, y, \sigma)}{2},$$

$$\frac{\partial^2 D}{\partial x^2} \approx D(x+1, y, \sigma) - 2D(x, y, \sigma) + D(x-1, y, \sigma),$$

and similarly for y , σ , and cross-terms. If $|\hat{x}_i| > 0.5$ in any dimension, the candidate is moved to the adjacent sample and the fit is repeated (up to 5 iterations). The interpolated value at the extremum is:

$$D(\hat{\mathbf{x}}) = D(\mathbf{x}_0) + \frac{1}{2} \frac{\partial D}{\partial \mathbf{x}} \hat{\mathbf{x}}.$$

- Low-contrast rejection.* Discard if $|D(\hat{\mathbf{x}})| < 0.03$ - the extremum is too weak to be stable under noise.
- Edge rejection.* Edges produce large DoG responses but are poorly localised along the edge direction. Detect this via the 2×2 spatial Hessian of D :

$$\mathbf{H} = \begin{bmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{bmatrix},$$

where $D_{xx} \approx D(x+1, y) - 2D(x, y) + D(x-1, y)$, etc. For a point on an edge, one eigenvalue $\lambda_1 \gg \lambda_2$. Instead of computing eigenvalues, use the identity $(\lambda_1 + \lambda_2)^2 / (\lambda_1 \lambda_2) = (\text{tr } \mathbf{H})^2 / \det \mathbf{H}$. Discard if

$$\frac{(\text{tr } \mathbf{H})^2}{\det \mathbf{H}} > \frac{(r+1)^2}{r}, \quad r = 10 \Rightarrow \text{threshold} = 12.1.$$

This ratio equals $(r+1)^2/r$ when $\lambda_1/\lambda_2 = r$ and is minimised (value 4) when $\lambda_1 = \lambda_2$.

3. Orientation assignment.

At the surviving keypoint's scale σ , compute gradient magnitude and orientation at each sample point in a circular neighbourhood of radius 3σ :

$$m(x, y) = \sqrt{[L(x+1, y) - L(x-1, y)]^2 + [L(x, y+1) - L(x, y-1)]^2},$$

$$\theta(x, y) = \tan^{-1} \left(\frac{L(x, y+1) - L(x, y-1)}{L(x+1, y) - L(x-1, y)} \right).$$

Each sample votes into a 36-bin histogram ($10^\circ/\text{bin}$) weighted by its magnitude and a Gaussian window of $\sigma_w = 1.5\sigma$:

$$h(\theta_k) += m(x, y) \cdot G(x - x_0, y - y_0, 1.5\sigma), \quad \theta_k = \text{bin containing } \theta(x, y).$$

The dominant peak is refined by fitting a parabola to the three bins around the peak:

$$\theta^* = \theta_{\text{peak}} - \frac{h(\theta_{\text{peak}} + 1) - h(\theta_{\text{peak}} - 1)}{2[h(\theta_{\text{peak}} + 1) - 2h(\theta_{\text{peak}}) + h(\theta_{\text{peak}} - 1)]} \cdot 10^\circ.$$

Any additional peaks within 80% of the maximum generate duplicate keypoints with those orientations. All subsequent operations rotate coordinates by $-\theta^*$, making them rotation-invariant.

4. Descriptor construction.

Work in a 16×16 pixel grid rotated to θ^* and sampled at scale σ (each sample is spaced σ apart, so the region spans 16σ pixels at the original resolution). Let (x', y') be coordinates in the rotated frame.

- Gradient computation.* At each of the $16 \times 16 = 256$ sample points, compute m and θ in $L(\cdot; \sigma)$ and rotate the orientation: $\theta' = \theta - \theta^*$.
- Gaussian weighting.* Each sample is weighted by a Gaussian centred on the keypoint with $\sigma_d = 0.5 \times 16 = 8$ (half the descriptor width in sample units):

$$w(x', y') = m(x', y') \cdot \exp\left(-\frac{x'^2 + y'^2}{2\sigma_d^2}\right).$$

- Histogram accumulation.* Divide the 16×16 grid into a 4×4 array of 4×4 -sample cells. Each cell accumulates an 8-bin orientation histogram ($45^\circ/\text{bin}$). Votes are distributed via *trilinear interpolation*: each sample contributes fractionally to the two adjacent spatial cells and the two adjacent orientation bins, weighted by proximity. This removes boundary effects from quantisation.
- Normalisation.*
 - Concatenate the $4 \times 4 \times 8 = 128$ histogram values: $\mathbf{f} \in \mathbb{R}^{128}$.
 - ℓ_2 -normalise: $\hat{\mathbf{f}} = \mathbf{f} / \|\mathbf{f}\|$.
 - Clamp: $\hat{f}_i \leftarrow \min(\hat{f}_i, 0.2)$.
 - Re-normalise: $\hat{\mathbf{f}} \leftarrow \hat{\mathbf{f}} / \|\hat{\mathbf{f}}\|$.

Step 3 limits the influence of any single gradient direction caused by non-linear illumination (e.g., specular highlights). Step 4 ensures unit norm after clamping.

Invariances and limits.

- Scale:* all computations use the characteristic scale σ - the physical region sampled is constant.
- Rotation:* orientations in the descriptor are relative to θ^* .
- Multiplicative illumination:* ℓ_2 -normalisation removes a global gain factor $I \rightarrow cI$.
- Saturated gradients:* clamping at 0.2 limits the impact of large gradient magnitudes.
- Not invariant to:* viewpoint changes beyond $\approx 60^\circ$ (perspective distortion exceeds the affine approximation), additive illumination (absolute intensity shifts), or spatially varying illumination.

Output. Running SIFT on a single image produces a **list of keypoints**, each carrying:

Field	Symbol	Meaning
Position	(x, y)	Sub-pixel location in image coordinates
Scale	σ	Characteristic scale (radius $\approx 3\sigma$ pixels)
Orientation	θ^*	Dominant gradient direction, $\theta^* \in [0^\circ, 360^\circ)$
Descriptor	$\hat{\mathbf{f}} \in \mathbb{R}^{128}$	Unit-norm 128-D gradient histogram vector

Together (x, y, σ, θ^*) define the *similarity frame* of the keypoint - a position, scale, and orientation that specifies exactly which image patch was described and in what canonical orientation. Two keypoints from different images of the same scene will produce the same (or nearly the same) $\hat{\mathbf{f}}$ if and only if their similarity frames cover the same physical region. Matching therefore proceeds by comparing the 128-D vectors; correspondence is declared when $\|\hat{\mathbf{f}}_A - \hat{\mathbf{f}}_B\|_2$ is small (subject to Lowe's ratio test). A typical image yields a few hundred to a few thousand keypoints.

RootSIFT (Arandjelović & Zisserman, 2012). Replace ℓ_2 normalisation with Hellinger (Bhattacharyya) normalisation: $f_i \leftarrow \sqrt{f_i} / \|\mathbf{f}\|_1$. Matching RootSIFT with ℓ_2 distance is equivalent to matching the original SIFT with the Hellinger kernel. This simple change reduces nearest-neighbour error rates by $\approx 20\%$ with zero extra cost.

5.2 SURF (Bay et al., 2006)

Speeded Up Robust Features approximates SIFT using box filters and integral images, achieving similar invariance at significantly lower cost.

Detection. Detect blobs via $\det \mathbf{H}_{\text{approx}}$, where the Gaussian second derivatives L_{xx} , L_{yy} , L_{xy} are approximated by box filters (computable in $O(1)$ via integral images). The approximation introduces a small constant correction:

$$\det \mathbf{H}_{\text{approx}} = D_{xx}D_{yy} - (0.9 D_{xy})^2.$$

Orientation assignment. Compute Haar wavelet responses d_x (horizontal) and d_y (vertical) at scale 4σ in a circular region of radius 6σ . Weight by a Gaussian centered on the keypoint. Slide a 60° wedge around the circle, summing d_x and d_y within it; the longest response vector gives θ^* .

Descriptor. Divide a $20\sigma \times 20\sigma$ patch (aligned to θ^*) into a 4×4 grid of $5\sigma \times 5\sigma$ sub-regions. In each sub-region compute Haar wavelet responses d'_x , d'_y (rotated to θ^*) at 25 sample points and form the 4-vector:

$$\mathbf{v}_k = \left(\sum d'_x, \sum d'_y, \sum |d'_x|, \sum |d'_y| \right).$$

Concatenate all $4 \times 4 \times 4 = 64$ values. The $|\cdot|$ components capture polarity of intensity change. Optionally extend to 128-D (U-SURF) by splitting positive and negative responses. ℓ_2 -normalise.

Comparison with SIFT. SURF is $\approx 3\times$ faster to detect and describe than SIFT. The 64-D descriptor is half the size. Both achieve similar matching performance; SURF is preferred when speed matters.

5.3 HOG - Dense Descriptor (Dalal & Triggs, 2005)

Histogram of Oriented Gradients is a *dense* descriptor designed for object detection (notably pedestrians), not sparse keypoint matching. It characterises the distribution of gradient orientations over a detection window.

Gradient computation. Compute $G_x = f * [-1, 0, 1]$ and $G_y = f * [-1, 0, 1]^T$. Magnitude $m = \sqrt{G_x^2 + G_y^2}$ and unsigned orientation $\theta = \arctan(G_y/G_x) \bmod 180^\circ$.

Cell histograms. Divide the detection window into non-overlapping 8×8 -pixel *cells*. In each cell, accumulate a 9-bin orientation histogram (0° – 180° , $20^\circ/\text{bin}$), weighting each pixel's vote by its gradient magnitude. The result per cell is a 9-D vector.

Block normalisation. Group 2×2 adjacent cells into overlapping 16×16 -pixel *blocks* (stride: one cell = 8 pixels). Concatenate the four cell histograms into a 36-D vector per block and ℓ_2 -normalise (L2-Hys: clip to 0.2, re-normalise). Overlapping blocks mean each cell's histogram appears in up to four normalised blocks, making the descriptor robust to local contrast changes.

Full descriptor. For a 64×128 detection window: $7 \times 15 = 105$ block positions (stride 8), each with a 36-D vector $\Rightarrow 105 \times 36 = 3780$ -D descriptor. Classify with a linear SVM.

Key design choices.

- Unsigned orientations (0–180°): sign of gradient is irrelevant for shape; symmetric gradients at dark-bright vs bright-dark edges map to the same bin.
- Overlapping block normalisation: compensates for spatially varying illumination better than a single global normalisation.
- Fine cells (8 × 8): preserves local spatial structure; coarser cells lose shape detail.

5.4 Binary Descriptors (BRIEF, ORB, and BRISK)

Binary descriptors replace floating-point vectors with bit strings, enabling Hamming-distance matching via XOR + popcount - orders of magnitude faster than ℓ_2 distance on floating-point vectors. Critical for mobile and embedded vision.

5.4.1 BRIEF (Calonder et al., 2010)

BRIEF. For a smoothed patch around the keypoint, sample n pairs of points (p_i, q_i) drawn from a Gaussian distribution. Each bit is:

$$b_i = \mathbf{1}[f(p_i) < f(q_i)].$$

Concatenate $n = 256$ bits into a 256-bit string. Extremely fast to compute and match. *Not* rotation-invariant - if the patch rotates, the fixed pairs (p_i, q_i) compare different pixel pairs, breaking the descriptor.

5.4.2 ORB (Rublee et al., 2011)

ORB. *Oriented FAST and Rotated BRIEF* - a patent-free, real-time alternative to SIFT/SURF. It bolts a rotation estimate (intensity centroid) onto FAST detection and fixes BRIEF’s rotation-sensitivity with a learned, steered bit-test pattern.

Algorithm.

1. *Oriented FAST - scale-aware keypoint detection.*
 - (a) Build a Gaussian pyramid of $L = 8$ levels, scale factor $s = 1.2$. Run FAST-9 independently at every level to get candidate keypoints at multiple scales.
 - (b) FAST assigns no corner *strength*. To enable NMS and select the N best keypoints (typically $N = 500$), compute the Harris response

$$R = \det M - k(\text{tr } M)^2$$

at each FAST candidate and discard non-maxima within a 7×7 neighbourhood. Harris is cheap to evaluate at pre-filtered candidates; computing it only at FAST survivors avoids scanning the whole image.

- (c) The scale level at which a keypoint is detected becomes its *characteristic scale*, giving partial scale invariance.
2. *Intensity centroid - fast orientation.* For each keypoint, compute image moments over a circular patch of radius r pixels:

$$m_{pq} = \sum_{x=-r}^r \sum_{y=-r}^r x^p y^q I(x, y), \quad p, q \in \{0, 1\}.$$

The *intensity centroid* is $C = (m_{10}/m_{00}, m_{01}/m_{00})$. The canonical orientation is the direction from the patch center to C :

$$\theta = \text{atan2}(m_{01}, m_{10}).$$

Why this works: in a corner patch, intensity is not symmetric - the centroid is displaced toward the heavier side. The vector from center to centroid points in a direction that is consistent across viewpoints, giving a repeatable canonical frame. Choosing r large enough (ORB uses $r = 15$ pixels at the detection scale) reduces noise. This is $O(r^2)$ - far cheaper than SIFT’s gradient histogram.

3. *Rotated BRIEF - steered, learned bit tests.*
Why naive steered BRIEF fails. A straight rotation of the BRIEF test pairs by θ (“steered BRIEF”) reduces correlation between the descriptor and the orientation angle, but the bit-pairs in the original BRIEF set were chosen randomly, so many pairs are correlated with each other - the effective dimensionality is much less than 256, hurting distinctiveness.

Learning uncorrelated, high-variance pairs. ORB selects the 256 pairs by a greedy algorithm on a training set of 300,000 patches:

- (a) Run all $\approx 10^5$ possible 31×31 -patch pixel-pair tests on every training patch.

- (b) For each test, compute the mean μ and variance σ^2 of the resulting bit across all patches. Tests near $\mu = 0.5$ have maximum variance (the bit is equally likely to be 0 or 1 - maximally informative).
 - (c) Sort tests by $|\mu - 0.5|$ (ascending). Start with the most uniform test.
 - (d) Greedily add the next test that has correlation < 0.6 with all already-selected tests.
 - (e) Stop after 256 tests are selected.
- This produces a set $\mathbf{S} = \{(p_i, q_i)\}_{i=1}^{256}$ of low-correlation, high-variance pairs.

Applying the rotation. Rather than rotating pairs by a continuous θ , ORB quantises θ into 30 bins (12° spacing) and precomputes a lookup table $\mathbf{S}_\theta$ for each bin - the pair coordinates rotated to that orientation. At description time, snap θ to the nearest bin and use $\mathbf{S}_\theta$ directly. This avoids floating-point rotation arithmetic at runtime. The bit string is then:

$$b_i = \mathbf{1}\left[I\left(p_i^{\mathbf{S}_\theta}\right) < I\left(q_i^{\mathbf{S}_\theta}\right)\right], \quad i = 1, \dots, 256.$$

Output. 256-bit descriptor, matched by Hamming distance $d_H(\mathbf{a}, \mathbf{b}) = \text{popcount}(\mathbf{a} \oplus \mathbf{b})$, computable in ≈ 5 CPU instructions per 64-bit word. Comparable distinctiveness to SIFT at $\approx 100\times$ the speed. Patent-free. Directly used in ORB-SLAM and ORB-SLAM2.

5.4.3 BRISK (Leutenegger et al., 2011)

BRISK. Samples $N = 512$ pairs from a fixed pattern of concentric rings around the keypoint. Long-range pairs (large spatial separation) contribute to orientation estimation; short-range pairs generate the binary descriptor. Scale is determined from the AGAST (a faster FAST variant) pyramid. 512-bit string; rotation- and scale-invariant.

Descriptor	Dim.	Type	Rot. inv.	Scale inv.	Speed
SIFT	128-D	float32	Yes	Yes	Slow
SURF	64-D	float32	Yes	Yes	Moderate
HOG	3780-D	float32	No	No	Moderate (dense)
BRIEF	256-bit	binary	No	No	Very fast
ORB	256-bit	binary	Yes	Partial	Very fast
BRISK	512-bit	binary	Yes	Yes	Fast

5.5 Feature Matching and Geometric Verification

Distance metrics. For float descriptors use ℓ_2 (Euclidean) distance; for binary descriptors use Hamming distance $d_H(\mathbf{a}, \mathbf{b}) = \text{popcount}(\mathbf{a} \oplus \mathbf{b})$ (number of differing bits).

Brute-force nearest neighbor. For each descriptor $\mathbf{d}$ in image A, find the closest descriptor in image B under the chosen metric. Cost: $O(n_A n_B D)$ for D -dimensional descriptors - quadratic in the number of keypoints.

Lowe’s ratio test. Let d_1 and d_2 be the distances to the nearest and second-nearest neighbor. Accept the match only if

$$\frac{d_1}{d_2} < 0.8.$$

Rationale: a correct match has a much closer neighbor than any incorrect alternative; an ambiguous match (near a repeated texture) has $d_1 \approx d_2$. This single test eliminates $\approx 90\%$ of false matches while retaining $\approx 95\%$ of correct ones (Lowe’s original figures).

Cross-check (mutual consistency). A match ($A_i \rightarrow B_j$) is accepted only if B_j also maps back to A_i as its nearest neighbor. Eliminates asymmetric false matches with no distance-ratio tuning. Can be combined with the ratio test.

Approximate nearest neighbor (ANN). Exact NN search in high dimensions is slow. FLANN (Fast Library for Approximate Nearest Neighbors) uses randomised k -d trees or hierarchical k -means trees to find approximate matches in $O(n \log n)$, with controllable accuracy/speed trade-off. Used in practice for large-scale matching.

RANSAC for geometric verification. Raw descriptor matching produces many outliers (incorrect correspondences). RANSAC (Random Sample And Consensus) robustly estimates a geometric model (homography H or fundamental matrix F) from noisy matches:

1. Randomly sample the minimum number of matches needed to fit the model (4 for H , 8 for F).
2. Fit the model to the sample.
3. Count *inliers*: matches whose reprojection error $\|\mathbf{x}' - H\mathbf{x}\| < \epsilon$ (typically $\epsilon = 3$ pixels).
4. Repeat for k iterations; keep the model with the most inliers.
5. Refit the model to all inliers.

Number of iterations needed: $k = \log(1-p)/\log(1-(1-e)^s)$, where p is the desired success probability (e.g., 0.99), e is the outlier fraction, and s is the sample size. For $e = 0.5$, $s = 4$: $k \approx 72$ iterations.

5.6 Affine Region Normalisation

Detectors like MSER produce *affine-covariant* regions - ellipses rather than circles - because the local structure may be stretched by viewpoint change. To apply a descriptor (designed for a canonical circular patch) to such a region, the ellipse must first be warped into a unit circle.

Ellipse from the second-moment matrix. At a detected region centre, compute the Gaussian-windowed structure tensor M (same as Harris). Its eigenvectors give the axes of the local intensity ellipse; the affine-normalised frame is obtained by the transformation

$$A = M^{1/2} = U\Lambda^{1/2}U^T,$$

where $M = U\Lambda U^T$ is the eigendecomposition. Applying A to the patch maps the ellipse to a unit circle.

Normalisation procedure.

1. Estimate M at the keypoint using a Gaussian window of scale σ .
2. Compute the affine map $A = M^{1/2}$ (or its inverse to warp the image patch).
3. Sample the image under the affine warp to produce a 41×41 canonical patch.
4. Assign orientation within the canonical patch (e.g., dominant gradient direction).
5. Compute any descriptor (SIFT, RootSIFT, etc.) on the canonical patch.

Iterating steps 1–2 (re-estimating M within the warped patch) refines the shape estimate; 3–5 iterations typically suffice. This procedure underlies the **Harris-Affine** and **Hessian-Affine** detectors - detectors that combine a standard interest-point response with iterative affine shape adaptation.

5.7 Image-Level Aggregation

The methods above produce a *set* of local descriptors per image - one per keypoint. Image-level tasks (retrieval, classification) require a *single fixed-length* representation of the whole image. The three classical approaches below aggregate the local descriptor set into such a vector.

5.8 Bag of Visual Words (BoW)

BoW is the classical method for scaling local features to image-level representation, enabling retrieval and classification over millions of images.

Visual vocabulary construction (offline).

1. Extract SIFT (or other) descriptors from a large training set - typically tens of millions of vectors.
2. Cluster them into K centroids using k -means (typical $K = 1,000$ – $1,000,000$). Each centroid is a *visual word*.
3. The K words form the **visual vocabulary** (codebook) $\mathcal{V} = \{\mathbf{c}_1, \dots, \mathbf{c}_K\}$.

Image representation (online).

1. Extract descriptors $\{\mathbf{d}_i\}$ from a query image.
2. *Quantise* each descriptor: assign it to its nearest visual word $v_i = \arg \min_k \|\mathbf{d}_i - \mathbf{c}_k\|$.
3. Build a K -dimensional histogram $\mathbf{h}$, where $h_k =$ count of descriptors assigned to word k .

4. Normalise $\mathbf{h}$ and apply **TF-IDF** weighting:

$$h_k \leftarrow h_k \cdot \underbrace{\log \frac{N}{N_k}}_{\text{IDF}_k},$$

where N is the total number of database images and N_k is the number that contain word k . Rare words (large IDF) receive higher weight; common words (appearing in nearly all images, likely background clutter) are down-weighted.

Image similarity is then the cosine distance between TF-IDF-weighted histograms.

Inverted index. For efficient retrieval, maintain a posting list per visual word: word k stores the list of all database images containing it. A query image touches only its $\sim K_q \ll K$ active words, so retrieval costs $O(K_q \cdot \bar{L})$ where $\bar{L}$ is the mean posting-list length - much faster than comparing the full histogram against every database image.

Limitations.

- Hard quantisation loses fine-grained distance information between nearby descriptors (two descriptors close to a Voronoi boundary are assigned to different words).
- Large K needed for discriminability; k -means on 10^7 vectors is expensive - hierarchical k -means (vocabulary tree, Nistér & Stewénus 2006) reduces assignment from $O(K)$ to $O(\log K)$.
- Spatial information is completely discarded - two images with identical word histograms but different spatial layouts score identically.

5.9 VLAD and Fisher Vector

Both aggregate local descriptors into a *fixed-length* image vector that is more compact and discriminative than a BoW histogram.

VLAD (Jégou et al., 2010). Vector of Locally Aggregated Descriptors. Given a vocabulary of K words $\{\mathbf{c}_k\}$, for each descriptor $\mathbf{d}_i$ assigned to word $k(i)$, accumulate the *residual* (the displacement from the centroid):

$$\mathbf{v}_k = \sum_{i: k(i)=k} (\mathbf{d}_i - \mathbf{c}_k).$$

Concatenate all K residual vectors into one vector $\mathbf{V} \in \mathbb{R}^{KD}$ (where D is the descriptor dimension). Apply per-column ℓ_2 normalisation, then power normalisation $\text{sign}(v)|v|^\alpha$ ($\alpha = 0.5$), then global ℓ_2 normalisation. For SIFT ($D = 128$) and $K = 64$: a $64 \times 128 = 8192$ -D vector - far smaller than a BoW histogram with $K = 10^6$.

VLAD captures *how* descriptors deviate from their word, not just *which* word they belong to, making it significantly more discriminative than BoW at the same K .

Fisher Vector (Perronnin & Dance, 2007). Models the descriptor distribution with a Gaussian Mixture Model (GMM) of K components fitted to training descriptors. For a new image, the Fisher Vector is the gradient of the log-likelihood w.r.t. the GMM parameters:

$$\mathbf{f} = \frac{1}{N} \sum_{i=1}^N \nabla_{\lambda} \log p(\mathbf{d}_i; \lambda).$$

This gradient has a closed form in terms of soft assignments γ_{ik} (posterior probability that $\mathbf{d}_i$ belongs to component k). Concatenating the gradients w.r.t. means and variances gives a $2KD$ -D vector. Fisher Vectors outperform VLAD and BoW for image classification with a linear SVM, and were state-of-the-art on ImageNet before deep learning.

5.10 Detector and Descriptor Evaluation

Detector evaluation - repeatability. Given a ground-truth homography H between two images, a keypoint (x_1, y_1) in image 1 is *repeated* if $H(x_1, y_1)$ falls within ϵ pixels of a keypoint in image 2 (typically $\epsilon = 3$). The **repeatability score** is the fraction of keypoints repeated, averaged over viewpoint/illumination changes. Standard benchmark: Oxford Affine dataset (Mikolajczyk et al., 2005).

Descriptor evaluation - matching score. Given repeated keypoints, a descriptor match is *correct* if it connects the same physical point. Standard metrics:

- **Precision–Recall:** threshold the descriptor distance; sweep the threshold to get the PR curve. A good descriptor has high area under the curve (AUC).
- **Matching score:** fraction of keypoints that are correctly matched (both repeated *and* correctly described) out of all repeated keypoints.
- **mAP** (mean Average Precision): for image retrieval, the mean of per-query AP over a query set. Standard on Oxford Buildings and Paris datasets.

HPatches benchmark (Balntas et al., 2017). A modern standard: 116 sequences, each with one reference patch and five transformed patches (57 illumination, 59 viewpoint). Evaluates descriptors on three tasks: patch verification (same/different?), image matching, and patch retrieval. Covers both floating-point and binary descriptors with unified metrics.

5.11 Learned Feature Descriptors

Hand-crafted descriptors (SIFT, ORB) encode domain knowledge about gradients and local structure. *Learned* descriptors replace this with representations optimised end-to-end from data, achieving substantially higher matching accuracy, especially under large viewpoint or illumination change.

Training objective. A descriptor network $f_\theta : \text{patch} \rightarrow \mathbb{R}^D$ is trained so that matching patches map to nearby vectors and non-matching patches map to distant vectors. Common losses:

- *Contrastive loss:* $\mathcal{L} = (1 - y)\frac{1}{2}d^2 + y\frac{1}{2}\max(0, m - d)^2$, where $d = \|f(\mathbf{p}) - f(\mathbf{q})\|$, $y = 1$ for non-matching pairs, and m is the margin.
- *Triplet loss:* $\mathcal{L} = \max(0, d(a, p) - d(a, n) + m)$ for anchor a , positive p , negative n . Directly enforces $d(a, p) < d(a, n) - m$.
- *Hardest-negative mining:* within a batch, select the hardest (most confusing) negative for each anchor. Critical for fast convergence.

Notable learned descriptors.

- **SuperPoint (DeTone et al., 2018):** a single homography-adapted network that detects keypoints *and* computes 256-D descriptors jointly. Trained via self-supervised homographic adaptation - synthetic transformations of MS-COCO images generate pseudo-ground-truth keypoints without manual annotation.
- **D2-Net (Dusmanu et al., 2019):** detects keypoints as local maxima of a dense CNN feature map, then uses the same map as the descriptor. The “detect-and-describe” approach ensures detection and description always agree on what counts as a salient point.
- **LoFTR (Sun et al., 2021):** abandons keypoint detection entirely. Uses a Transformer with cross-attention to match dense feature maps between two images directly, producing semi-dense correspondences. Particularly robust in textureless regions where keypoint detectors fail.

Key trend. Learned detectors and descriptors consistently outperform hand-crafted ones on standard benchmarks (HPatches, outdoor localisation). The remaining advantage of classical methods (SIFT, ORB) is speed, interpretability, and zero training data requirements.